

UIU Lost & Found

Database Design

Overview

This document describes the complete database schema for the UIU Lost & Found web application. The database consists of 10 tables covering all features of the platform — from user authentication and post management to claims, messaging, notifications, and moderation.

The two central tables are **USERS** and **POSTS**. All other tables relate back to one or both of these.

1. USERS

Stores all registered students and admins. Every action on the platform is tied to a user account.

USERS		
Column	Type	Constraint
id	INT	Primary Key, Auto Increment
full_name	VARCHAR(100)	Not Null
email	VARCHAR(100)	Not Null, Unique
password_hash	VARCHAR(255)	Not Null
student_id	VARCHAR(20)	Nullable
phone	VARCHAR(20)	Nullable
role	ENUM	Values: user, admin — Default: user
created_at	TIMESTAMP	Default: Current Timestamp

Used by: Login, Register, Dashboard, All pages

2. POSTS

The core table. Each row is either a lost or found item report submitted by a user. The 'type' field distinguishes between lost and found posts. 'status' tracks whether the item has been resolved.

POSTS		
Column	Type	Constraint
id	INT	Primary Key, Auto Increment
user_id	INT	Foreign Key -> USERS(id)
type	ENUM	Values: lost, found — Not Null

POSTS		
title	VARCHAR(150)	Not Null
category	INT	Foreign Key -> CATEGORIES(id)
description	TEXT	Not Null
location	INT	Foreign Key -> LOCATIONS(id)
specific_spot	VARCHAR(150)	Nullable
date_occurred	DATE	Not Null
time_occurred	TIME	Nullable
status	ENUM	Values: active, resolved, removed — Default: active
held_at	VARCHAR(100)	Nullable (for found posts only)
created_at	TIMESTAMP	Default: Current Timestamp

Used by: Report Lost, I Found Something, Browse Listings, Item Detail, Dashboard, Homepage

3. IMAGES

Stores photo uploads for each post. Kept separate from POSTS so one post can have multiple images.

IMAGES		
Column	Type	Constraint
id	INT	Primary Key, Auto Increment
post_id	INT	Foreign Key -> POSTS(id)
image_url	VARCHAR(255)	Not Null
uploaded_at	TIMESTAMP	Default: Current Timestamp

Used by: Report Lost, I Found Something, Browse Listings, Item Detail

4. CATEGORIES

Stores item categories such as Electronics, Keys, ID/Cards, etc. as proper database records instead of hardcoded values in the frontend.

CATEGORIES		
Column	Type	Constraint
id	INT	Primary Key, Auto Increment
name	VARCHAR(50)	Not Null, Unique

CATEGORIES		
icon	VARCHAR(10)	Emoji icon for UI display

Used by: Post forms, Browse Listings (filter)

5. LOCATIONS

Stores UIU campus locations such as Library, Cafeteria, Lab, etc. Allows consistent filtering and searching by location.

LOCATIONS		
Column	Type	Constraint
id	INT	Primary Key, Auto Increment
name	VARCHAR(100)	Not Null, Unique
area	VARCHAR(100)	Nullable (building/floor)

Used by: Post forms, Browse Listings (filter)

6. CLAIMS

When a user believes a found item belongs to them, they submit a claim. The post owner can approve or reject the claim. Status tracks the outcome.

CLAIMS		
Column	Type	Constraint
id	INT	Primary Key, Auto Increment
post_id	INT	Foreign Key -> POSTS(id)
claimant_id	INT	Foreign Key -> USERS(id)
message	TEXT	Claimant's ownership description
status	ENUM	Values: pending, approved, rejected — Default: pending
created_at	TIMESTAMP	Default: Current Timestamp

Used by: Item Detail ('This is mine!' button), Dashboard

7. MESSAGES

Enables private in-site messaging between users about a specific post, so contact details are not exposed publicly.

MESSAGES		
Column	Type	Constraint
id	INT	Primary Key, Auto Increment
sender_id	INT	Foreign Key -> USERS(id)
receiver_id	INT	Foreign Key -> USERS(id)
post_id	INT	Foreign Key -> POSTS(id)
content	TEXT	Not Null
is_read	BOOLEAN	Default: False
created_at	TIMESTAMP	Default: Current Timestamp

Used by: Item Detail (contact button), Dashboard (inbox)

8. NOTIFICATIONS

Stores alerts for users — for example when someone claims their post, sends a message, or when their post is resolved.

NOTIFICATIONS		
Column	Type	Constraint
id	INT	Primary Key, Auto Increment
user_id	INT	Foreign Key -> USERS(id)
post_id	INT	Foreign Key -> POSTS(id)
type	VARCHAR(50)	e.g. 'claim', 'message', 'resolved'
message	VARCHAR(255)	Notification text
is_read	BOOLEAN	Default: False
created_at	TIMESTAMP	Default: Current Timestamp

Used by: Dashboard (notification bell), Navbar

9. REPORTS

Allows users to flag suspicious or inappropriate posts. Admins review reports through the admin panel and can remove posts.

REPORTS		
Column	Type	Constraint
id	INT	Primary Key, Auto Increment

REPORTS		
post_id	INT	Foreign Key -> POSTS(id)
reporter_id	INT	Foreign Key -> USERS(id)
reason	TEXT	Not Null
status	ENUM	Values: pending, reviewed, dismissed — Default: pending
created_at	TIMESTAMP	Default: Current Timestamp

Used by: Item Detail (report button), Admin Panel

10. SUCCESS_STORIES

Stores stories from users who successfully recovered their lost items. Displayed on the homepage to build community trust.

SUCCESS_STORIES		
Column	Type	Constraint
id	INT	Primary Key, Auto Increment
post_id	INT	Foreign Key -> POSTS(id)
user_id	INT	Foreign Key -> USERS(id)
story	TEXT	Not Null
created_at	TIMESTAMP	Default: Current Timestamp

Used by: Homepage (success stories section)

Relationships Summary

From	To	Relationship	Meaning
USERS	POSTS	One to Many	A user creates many posts
USERS	CLAIMS	One to Many	A user submits many claims
USERS	MESSAGES	One to Many	A user sends many messages
USERS	NOTIFICATIONS	One to Many	A user receives many notifications
USERS	REPORTS	One to Many	A user files many reports
USERS	SUCCESS_STORIES	One to Many	A user shares many stories
POSTS	IMAGES	One to Many	A post has many images
POSTS	CLAIMS	One to Many	A post receives many claims

POSTS	MESSAGES	One to Many	A post has many messages
POSTS	NOTIFICATIONS	One to Many	A post triggers many notifications
POSTS	REPORTS	One to Many	A post can be flagged many times
POSTS	SUCCESS_STORIES	One to Many	A post can result in a story
CATEGORIES	POSTS	One to Many	A category classifies many posts
LOCATIONS	POSTS	One to Many	A location is used by many posts

Pages & Tables Used

Page	Tables Used
Homepage	POSTS, IMAGES, SUCCESS_STORIES
Login	USERS
Register	USERS
Report Lost Item	POSTS, IMAGES, CATEGORIES, LOCATIONS, USERS
I Found Something	POSTS, IMAGES, CATEGORIES, LOCATIONS, USERS
Browse Listings	POSTS, IMAGES, CATEGORIES, LOCATIONS
Item Detail	POSTS, IMAGES, USERS, CLAIMS, MESSAGES, REPORTS
User Dashboard	USERS, POSTS, IMAGES, CLAIMS, MESSAGES, NOTIFICATIONS
About / FAQ	None (static page)
Admin Panel	POSTS, USERS, REPORTS, CLAIMS